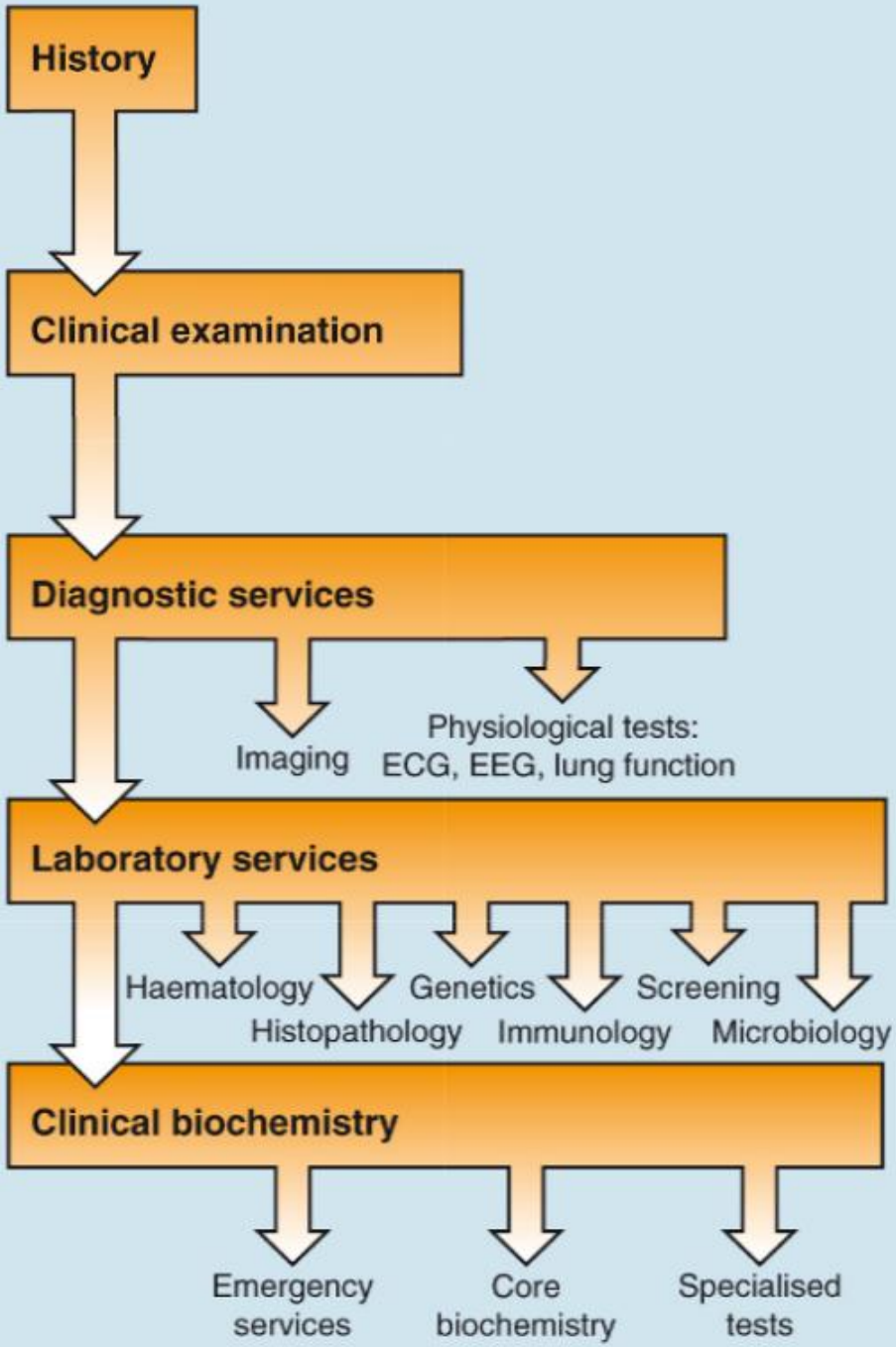
Clínica Universidad de Navarra

Principles of Laboratory Medicine

Dr. Nerea Varo
Clinical Biochemistry Laboratory
Clínica Universidad de Navarra

Clinical decision



Laboratory Medicine

The clinical laboratory in patient care

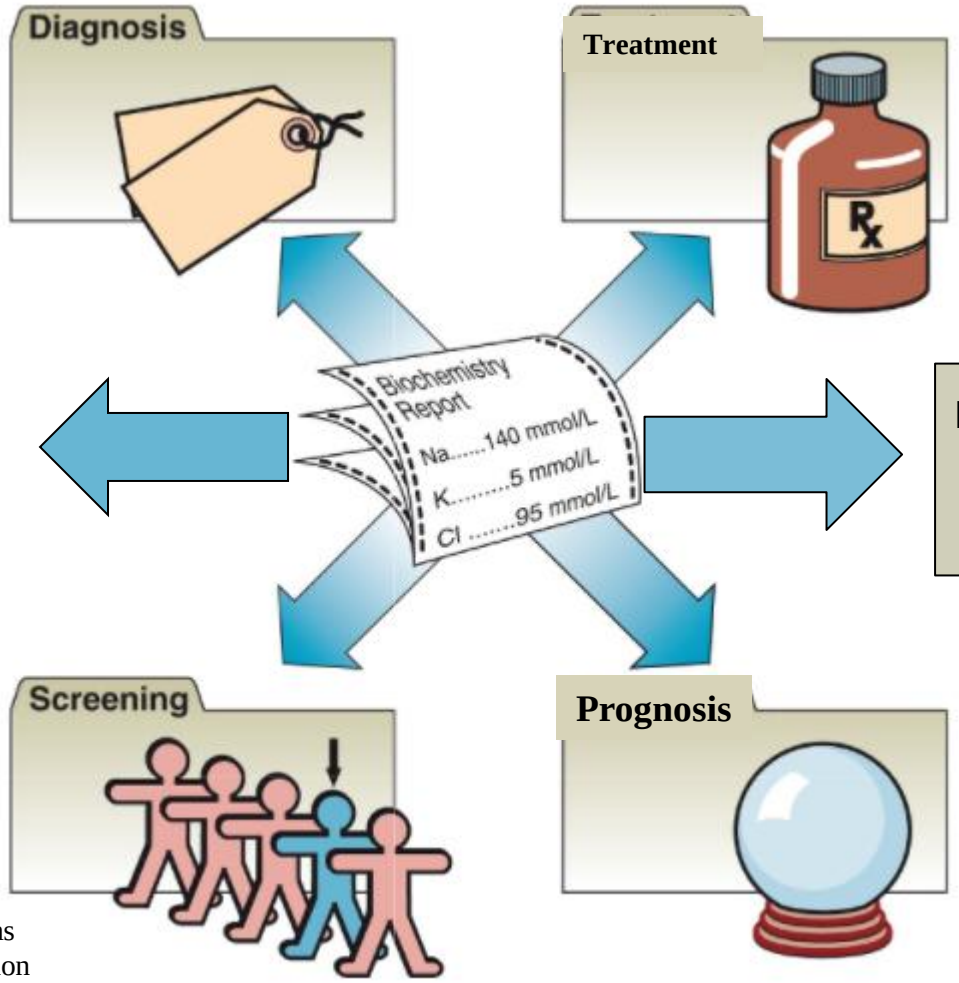
Purpose of the Clinical Laboratory

Facility in a medical setting where human samples are analysed to obtain reliable and reproducible analytical results that provide the clinician an aid in screening, diagnosing or monitoring patients

Analytical process

Preanalytical → Analitical → Postanalitical →

The use of biochemical tests…



confirm a disease

Choose the treatment
Measure the drug
Tumors
Hepatitis

Athletes
Occupational Medicine
Primary Care

No symptoms
Early detection
Colon cancer
Congenital hypothyroidism

Tumor markers
Troponin

The analytical process

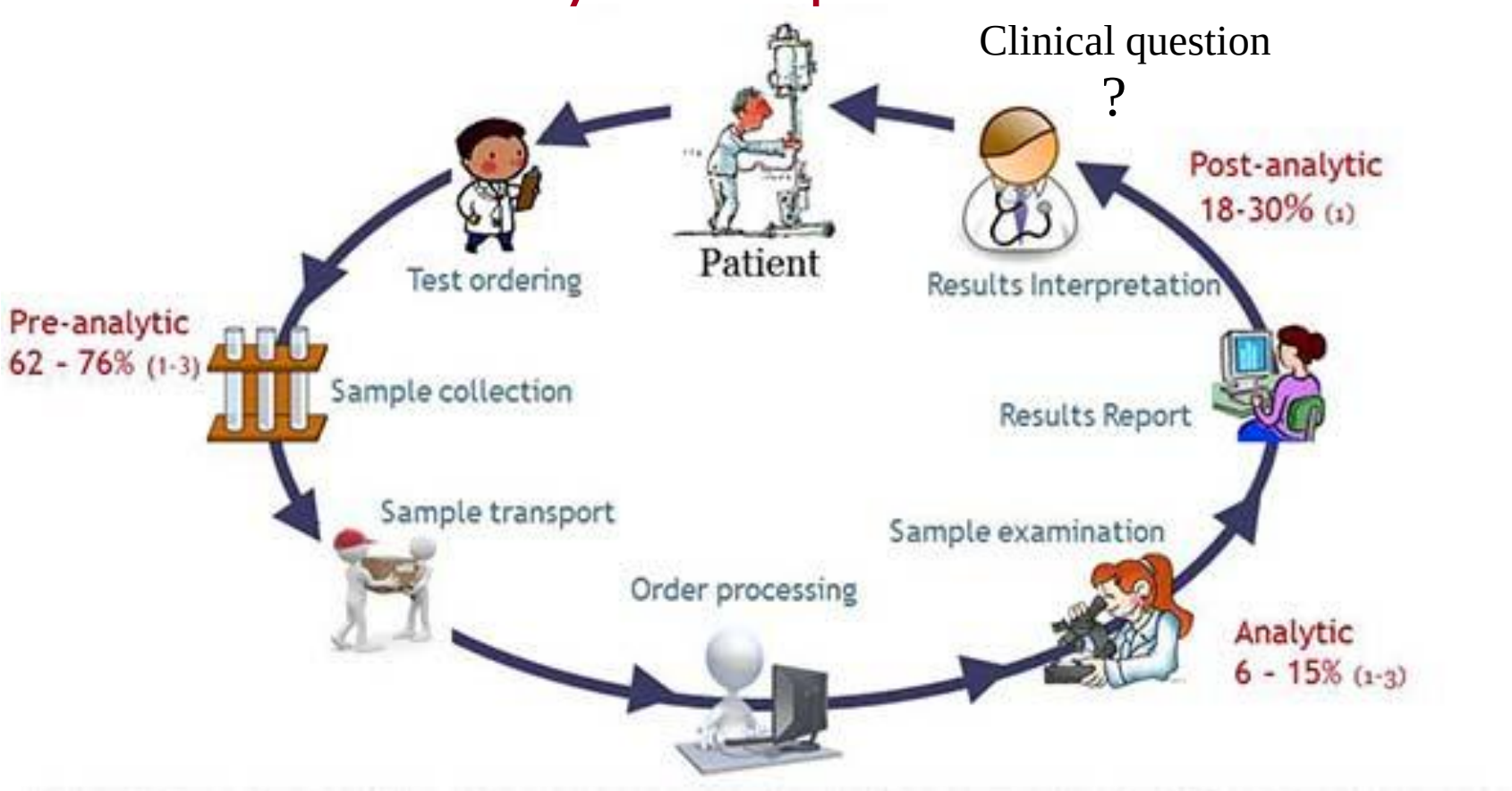
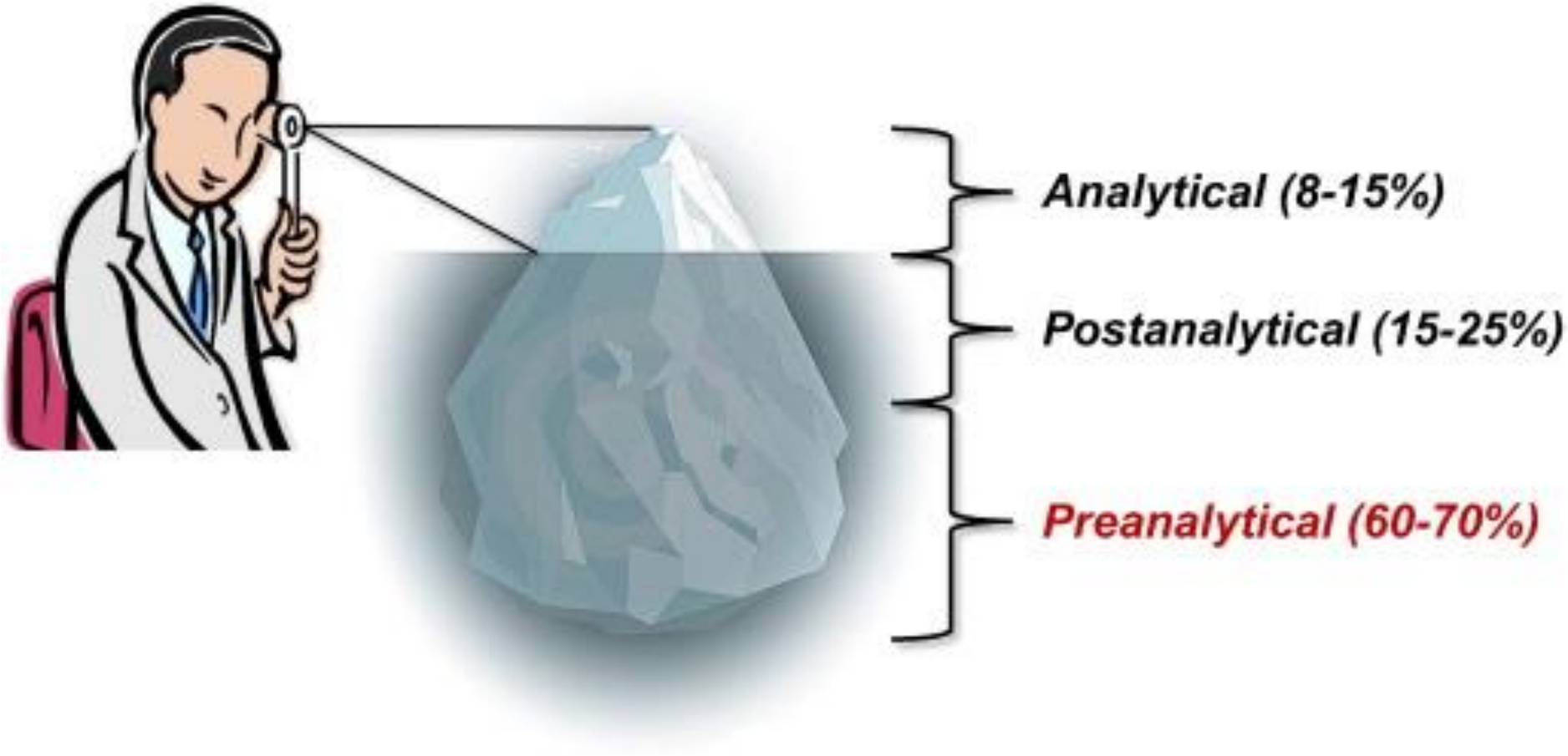
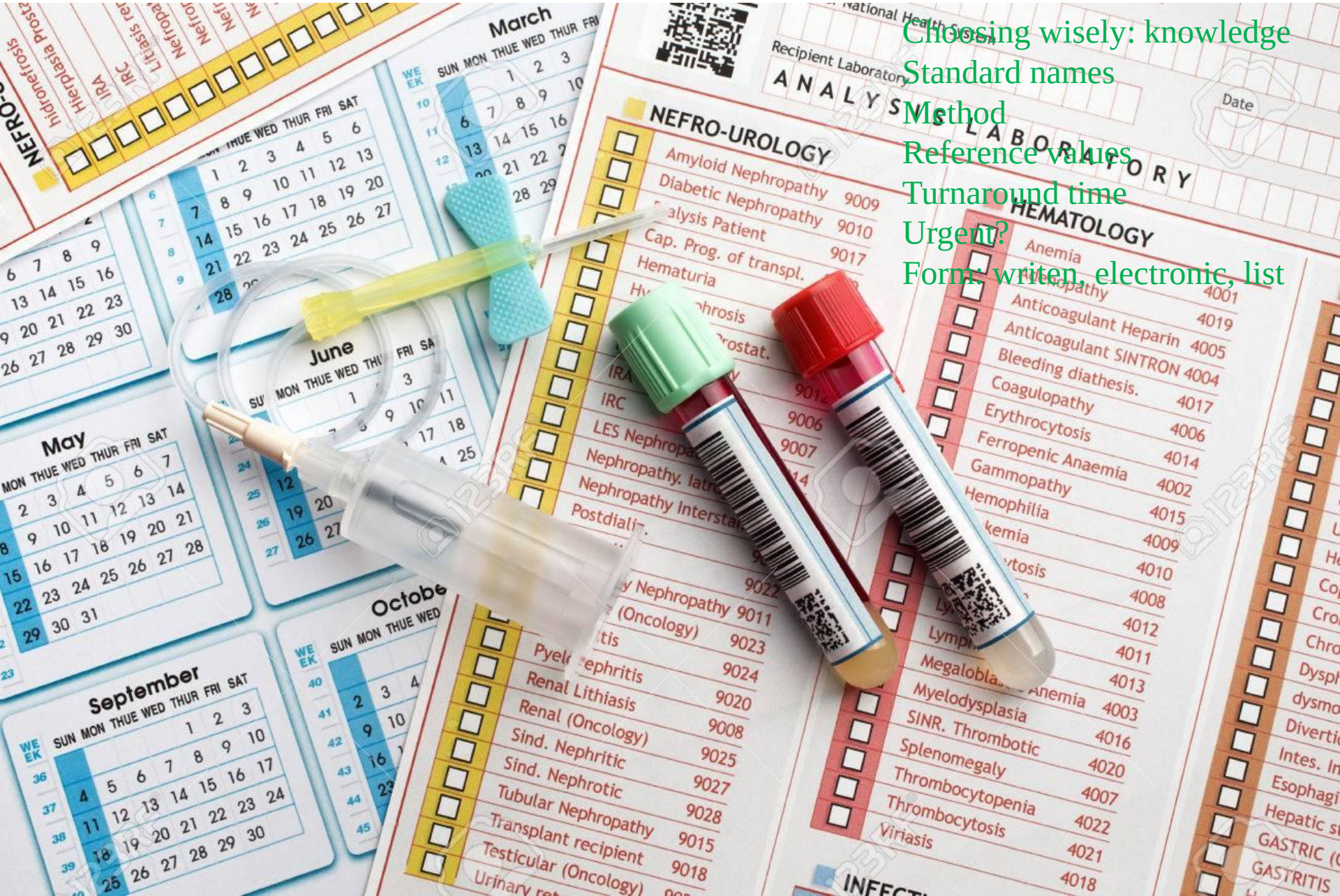


Figure 1. Total Laboratory Testing Process. In red: percentage of errors reported in that phase according to: (1) Restelli V., Taylor A., Cochrane D., Noble M. (2017) Medical laboratory associated errors: the 33-month experience of an on-line volunteer Canadian province wide error reporting system. Diagnosis (aop); (2) Carraro, P. (2007) Errors in a Stat Laboratory: Types and Frequencies 10 Years Later. Clin. Chem. 53(7) (3) Plebani, M. (2006) Errors in clinical laboratories or errors in laboratory medicine? Clin. Chem. Lab. Med. 44 (6)750-759

Errors in the analytical process



Test catalog and ordering



Test catalog and ordering

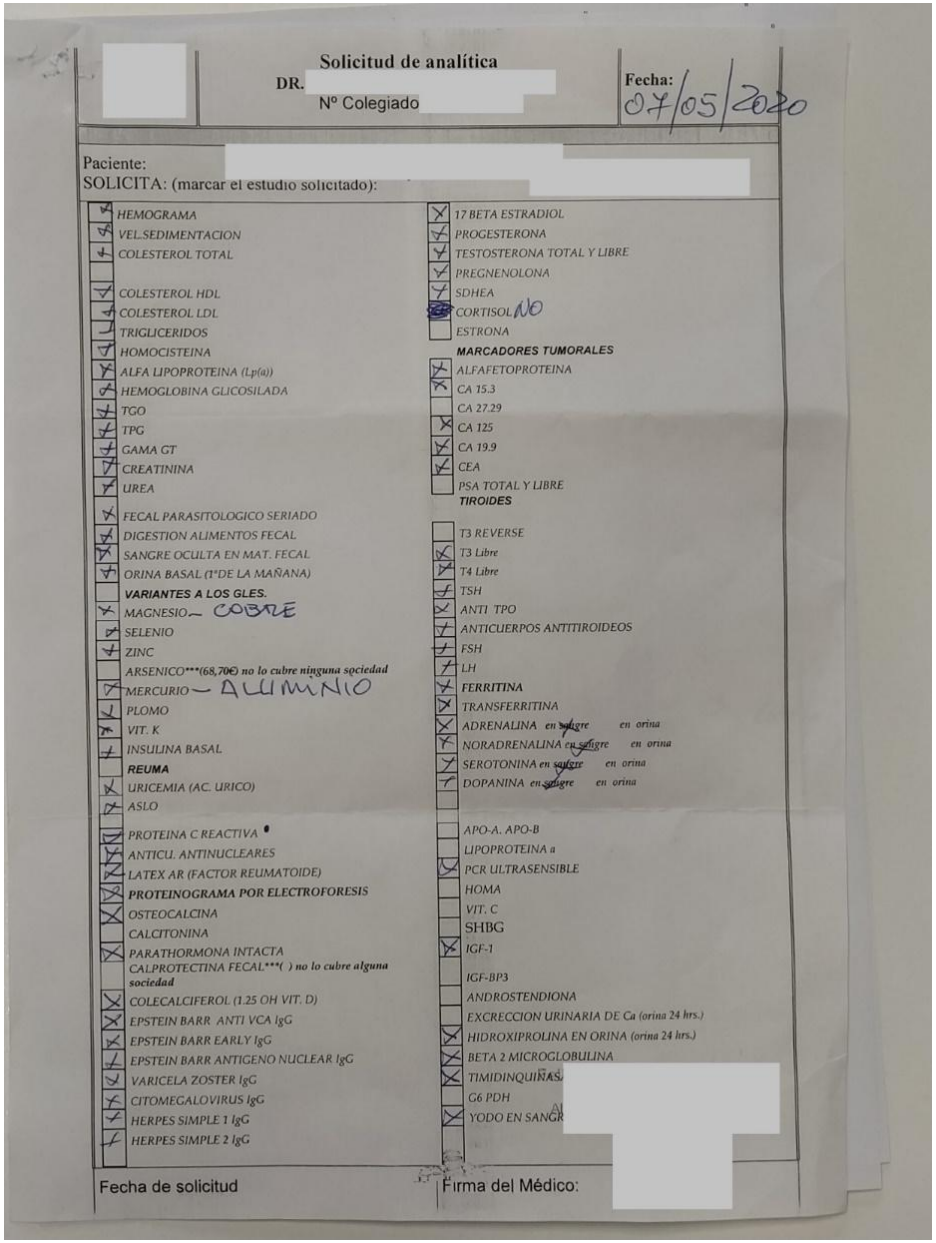
?

think

minimise

protocols

Clinical information



Solicitud de analítica

DR. [redacted]
Nº Colegiado

Fecha: 07/05/2020

Paciente: [redacted]
SOLICITA: (marcar el estudio solicitado):

- HEMOGRAMA
- VEL.SEDIMENTACION
- COLESTEROL TOTAL
- COLESTEROL HDL
- COLESTEROL LDL
- TRIGLICERIDOS
- HOMOCISTEINA
- ALFA LIPOPROTEINA (Lp(a))
- HEMOGLOBINA GLICOSILADA
- TGO
- TPG
- GAMA GT
- CREATININA
- UREA
- FECAL PARASITOLOGICO SERIADO
- DIGESTION ALIMENTOS FECAL
- SANGRE OCULTA EN MAT. FECAL
- ORINA BASAL (1ª DE LA MAÑANA)

VARIANTES A LOS GLES.

- MAGNESIO — COBRE
- SELENIO
- ZINC
- ARSENICO***(68,70€) *no lo cubre ninguna sociedad*
- MERCURIO — ALUMINIO
- PLOMO
- VIT. K
- INSULINA BASAL

REUMA

- URICEMIA (AC. URICO)
- ASLO
- PROTEINA C REACTIVA
- ANTICU. ANTINUCLEARES
- LATEX AR (FACTOR REUMATOIDE)

PROTEINOGRAMA POR ELECTROFORESIS

- OSTEOCALCINA
- CALCITONINA
- PARATHORMONA INTACTA
- CALPROTECTINA FECAL***( ) *no lo cubre alguna sociedad*
- COLECALCIFEROL (1.25 OH VIT. D)
- EPSTEIN BARR ANTI VCA IgG
- EPSTEIN BARR EARLY IgG
- EPSTEIN BARR ANTIGENO NUCLEAR IgG
- VARICELA ZOSTER IgG
- CITOMEGALOVIRUS IgG
- HERPES SIMPLE 1 IgG
- HERPES SIMPLE 2 IgG

- 17 BETA ESTRADIOL
- PROGESTERONA
- TESTOSTERONA TOTAL Y LIBRE
- PREGNENOLONA
- SDHEA
- CORTISOL NO
- ESTRONA

MARCADORES TUMORALES

- ALFAFETOPROTEINA
- CA 15.3
- CA 27.29
- CA 125
- CA 19.9
- CEA
- PSA TOTAL Y LIBRE

TIROIDES

- T3 REVERSE
- T3 Libre
- T4 Libre
- TSH
- ANTI TPO
- ANTICUERPOS ANTITIROIDEOS
- FSH
- LH
- FERRITINA
- TRANSFERRITINA
- ADRENALINA en sangre en orina
- NORADRENALINA en sangre en orina
- SEROTONINA en sangre en orina
- DOPANINA en sangre en orina
- APO-A. APO-B
- LIPOPROTEINA a
- PCR ULTRASENSIBLE
- HOMA
- VIT. C
- SHBG
- IGF-1
- IGF-BP3
- ANDROSTENDIONA
- EXCRECCION URINARIA DE Ca (orina 24 hrs.)
- HIDROXIPROLINA EN ORINA (orina 24 hrs.)
- BETA 2 MICROGLOBULINA
- TIMIDINQUINASA
- G6 PDH
- YODO EN SANGRE

Fecha de solicitud

Firma del Médico:

Patient identification: patient´s safety



Sentinel indicator
Prevent errors

phlebotomist

verbalize

hospital inpatients
Wrist bands

discrepancy

Two unique identifiers

Active questions

Specimen for clinical laboratory tests

Sample + information

TABLE 2.1 Specimens used for biochemical analyses

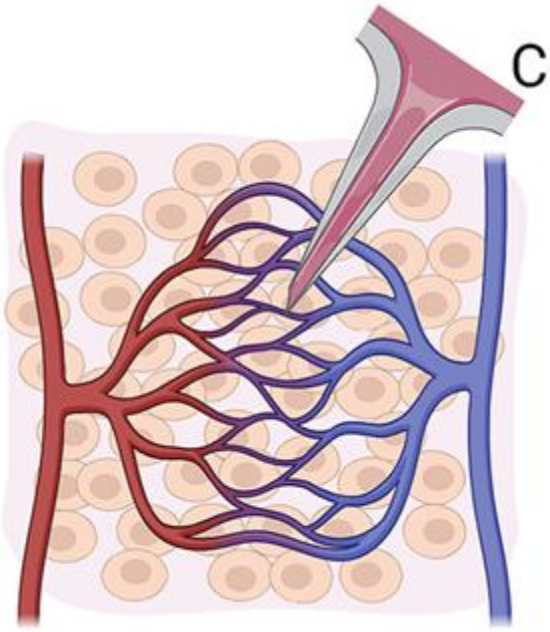
- Venous blood, serum or plasma
- Arterial blood
- Capillary blood
- Blood spot on a filter paper (Guthrie card)
- Urine
- Faeces
- Cerebrospinal fluid (CSF)
- Saliva
- Tissue and cells
- Aspirates, e.g.
  - pleural fluid
  - ascites
  - joint (synovial) fluid
  - intestinal (duodenal)
  - pancreatic pseudocysts
- Calculi (stones)

- Breath

Blood collection sites in human body

Type	Collection site	Common use
VENOUS	Arm veins	Lab tests
ARTERIAL	Arteries	Blood gas análisis
CAPILLARY Liquid Dried blood spots	Finger/Heel	 Diabetes monitoring New born screening

Capillary blood



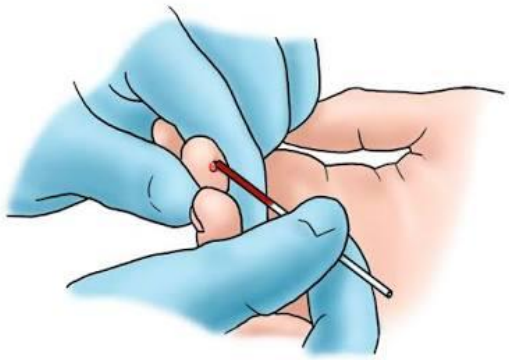
Capillary blood:

- Venous blood
- Arterial blood
- Interstitial fluid
- Intracellular fluid

Capillary blood in core laboratories: current and future challenges

Álvaro González, Sara Deza, Julia Maroto-García and Nerea Varo

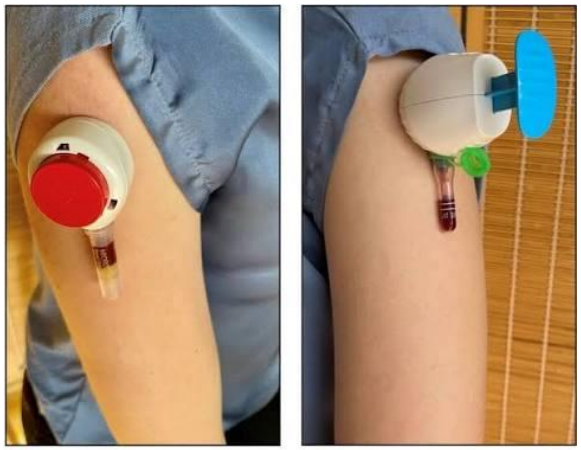
Capillary blood collection



Liquid capillary blood



Dried blood spot card



Upper arm capillary blood

Blood specimens

TBL 6.1



TABLA 1-1 Tipos de tubos de muestra que se emplean habitualmente en el laboratorio

	Anticoagulante	Color del tapón	Ejemplos de determinaciones
Suero		Marrón	Enzimas, glucosa, lípidos y otros Iones: Ca^{2+}, Mg^{2+}, Fe^{3+} y Li^{+} Serología: de hepatitis, del VIH, etc. Hormonas, marcadores tumorales y otros
Plasma	EDTA-K_3	Lila	Hormonas Pruebas de urgencias
	Heparina de litio	Verde	Hormonas Electrolitos: Na^{+}, K^{+}, Cl^{-} y HCO_3^{-}
	Fluoruro/oxalato	Gris	Lactato
	Citrato de sodio	Azul	Pruebas de coagulación Velocidad de sedimentación
Sangre	EDTA-K_3	Lila	Glicohemoglobina A1c Hemograma

Anticoagulants

- EDTA: calcium quelator (irreversible)
- Lithium heparin: inhibits trombin
- Citrate: calcium quelator (reversible)

Conservantes

- Sodium fluoride: inhibits glucolisis (lactate)
- Aprotinin: proteases inhibitor

Serum obtention

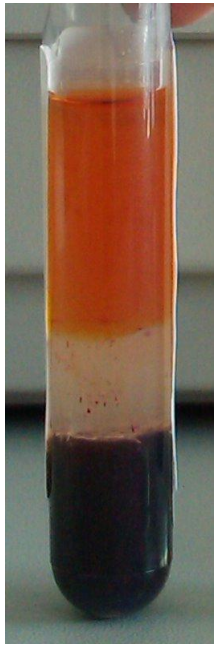
TBL 6.1

Non centrifuged specimen

Centrifuged (spun down) specimen



Separating gel



Serum 55-60%

Red cells 40-45%

TBL 6.1

Urine and faeces containers

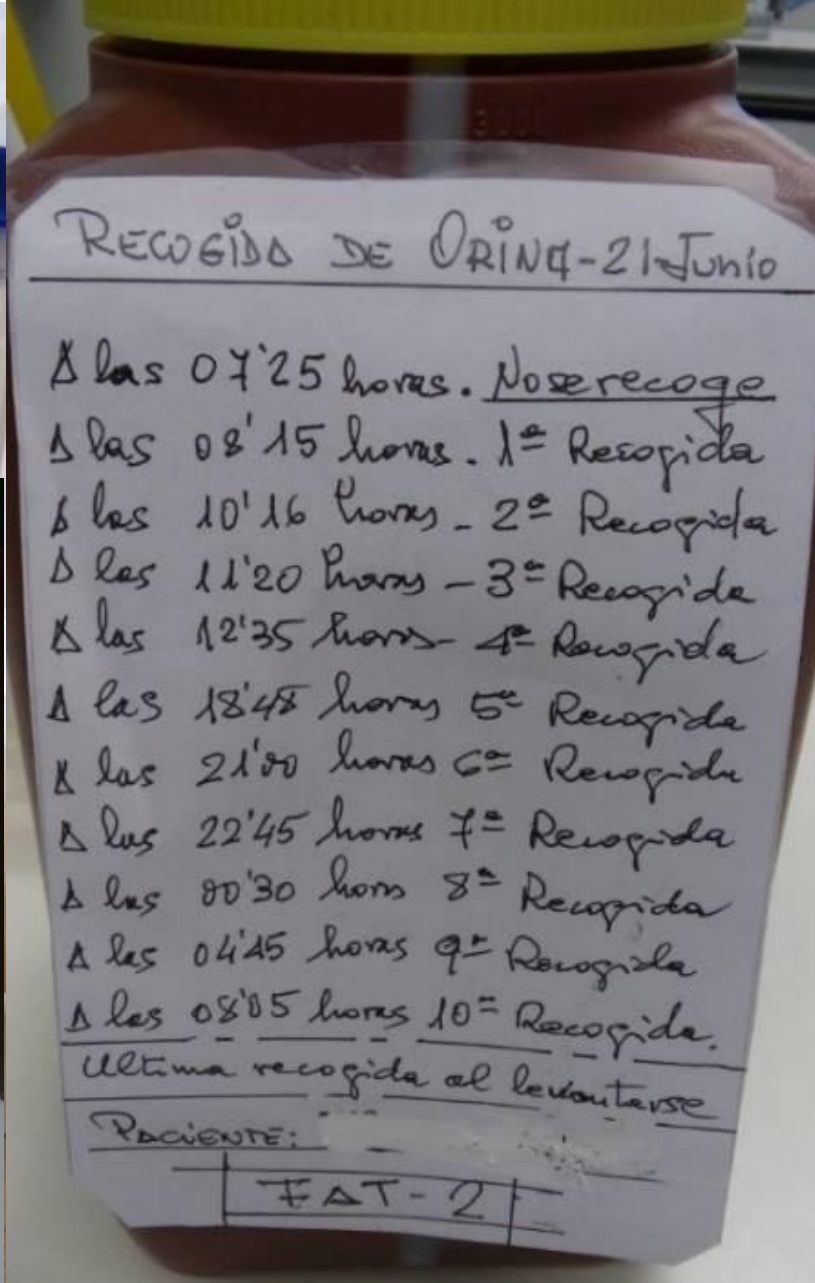


Capacity

preservative







I've seen things you people wouldn't believe.
(Blade Runner)

Breath tests



Breath tests

Exhaled air

Bags

Non-invasive

Stable

Portable



Gastrointestinal: lactose intolerance, SIBO (Small Intestinal Bacterial Overgrowth)

Preanalytical Variability
Non-modifiable Biological Variability

Patient-specific factors that must be interpreted in context but cannot be standardized before sample collection.

Sex

Testosterone and estradiol show different concentrations in men and women; some muscle enzymes may also vary.

Age

Alkaline phosphatase may be higher during growth due to increased osteoblastic activity.

Race / ancestry

Some analytes show population differences; for example, PSA levels may be higher in Black men.

Pregnancy

Physiological changes during pregnancy affect multiple parameters, including increased cholesterol concentrations.



Key message: these factors cannot be “corrected”; they must be recognized to ensure appropriate interpretation of results.

Preanalytical Variability
Modifiable Biological Variability

Habits and conditions that may significantly alter the concentration of specific analytes.

Diet / food intake

Recent food intake may increase triglyceride and glucose concentrations.

Biological rhythms

Circadian: cortisol · Monthly: sex hormones · Seasonal: vitamin D.

Stress

May increase cortisol and catecholamine levels. When appropriate, alternative matrices such as saliva or urine may be considered.

Physical exercise

Exercise may increase muscle biomarkers such as creatine kinase (CK).

Medications

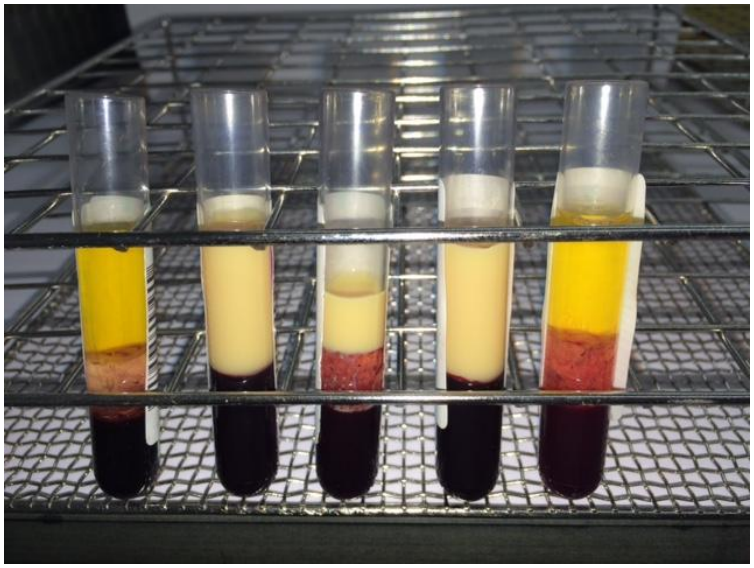
When clinically feasible, the timing of testing in relation to treatment should be considered. In chronic therapies, medication is part of the patient's baseline condition.



How does recent intake afect test results?

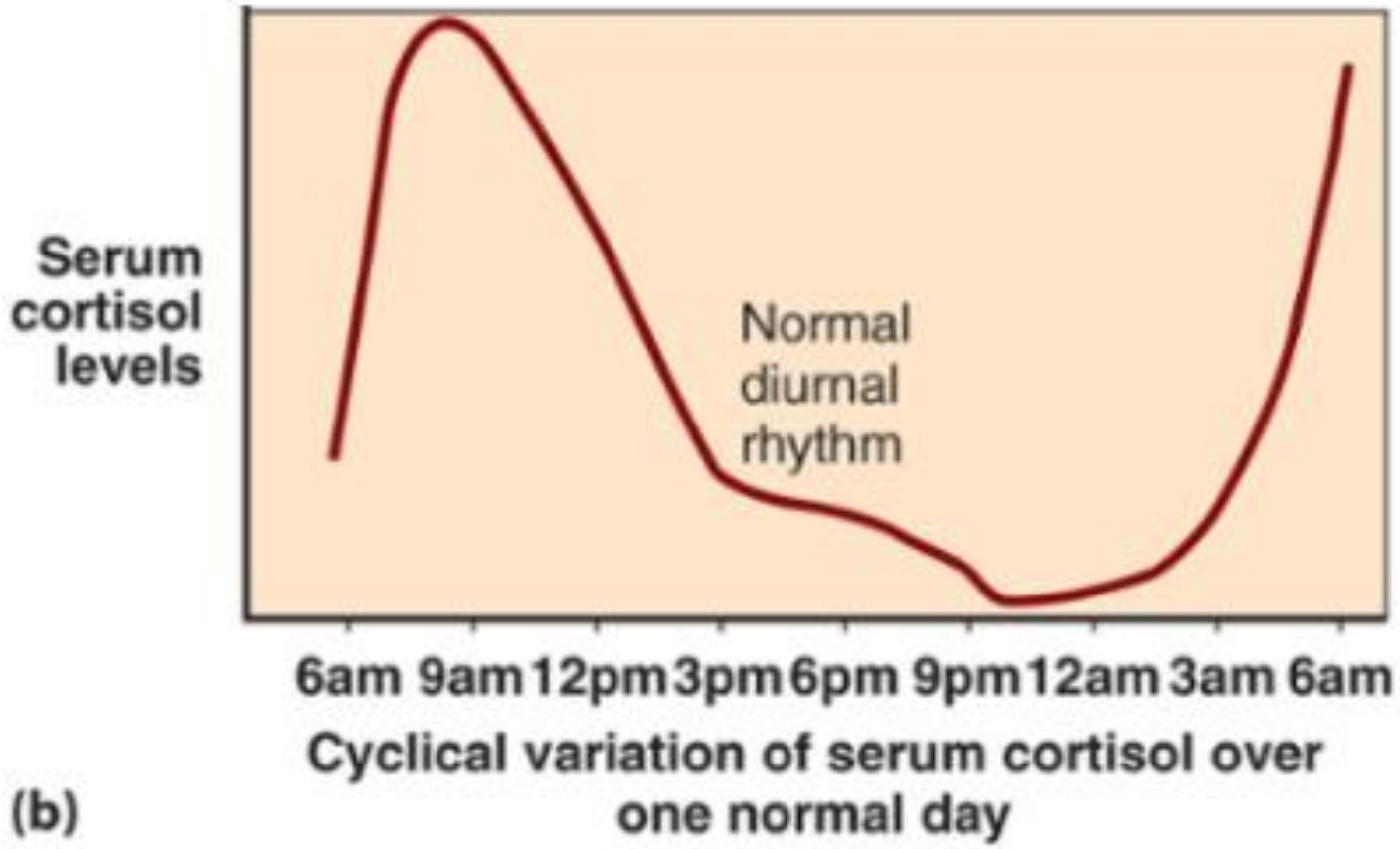
1. Enhanced analyte
   - glucose
   - triglycerides

2. Metabolic changes
   - glucagon / insulin
   - Growth hormone
   - Protein diet: urea and uric acid

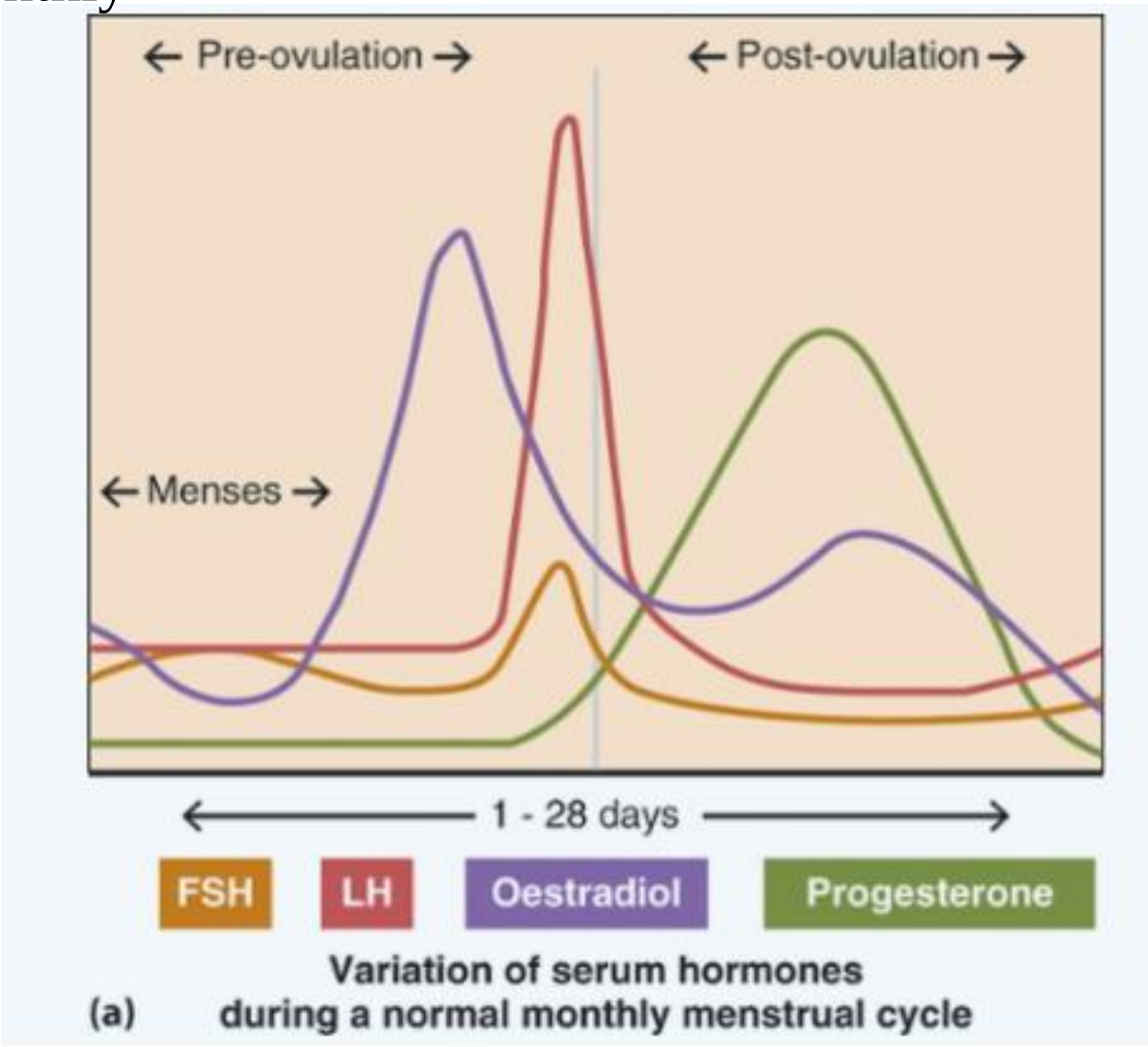


3. Lipemia: interference
   - Turbidimetric and spectrophotometric methods
   - Aqueous component displacement
   - Decreases antibodies accessibility in immunoassays

circadian

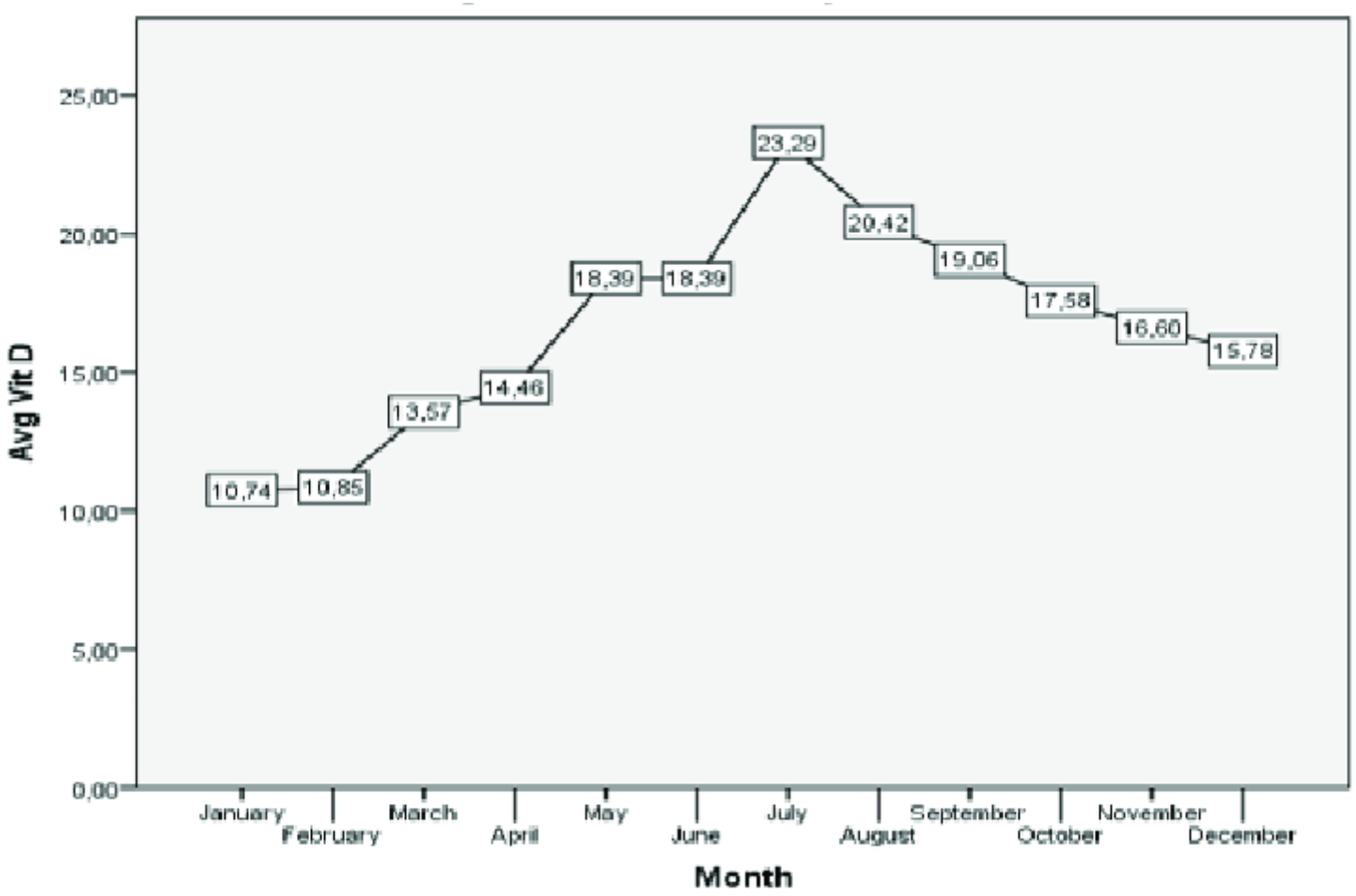

Serum cortisol levels
Normal diurnal rhythm
6am 9am 12pm 3pm 6pm 9pm 12am 3am 6am
Cyclical variation of serum cortisol over one normal day
(b)

Monthly



(a) Variation of serum hormones during a normal monthly menstrual cycle

Seasonal variation. Vitamin D by months



Evaluation of seasonal variation of vitamin D levels according to demographic criteria. Annals of Medical Research 2020

Preanalytical Variability
Blood Collection: Standardization to Reduce Errors

A substantial proportion of preanalytical variability arises during sample collection and initial handling.

Body position

Sitting, standing, or lying down may alter the distribution of water and solutes. In hospitalized patients, a consistent body position is preferable.

Tourniquet

Use for the shortest possible duration. Prolonged stasis promotes local changes and anaerobic metabolism; lactate is a sensitive example.

Collection tube and handling

Use the appropriate collection tube and follow stability, transport, and storage requirements for each analyte.

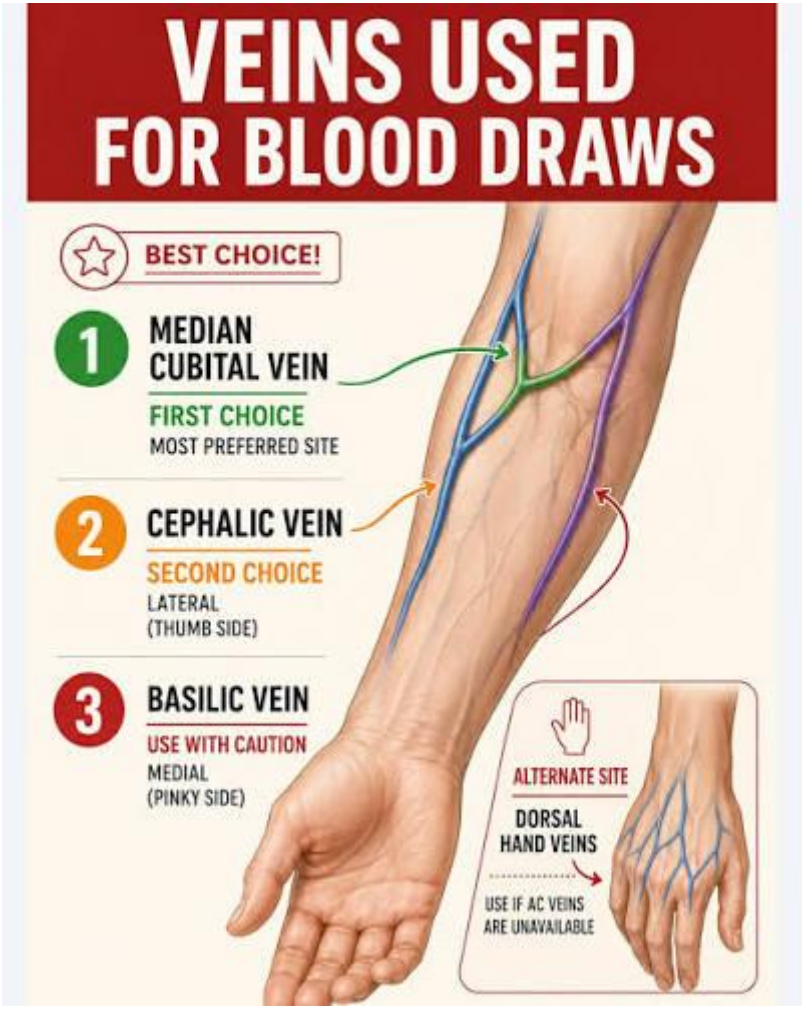
Sample characteristics

Review visible or instrument-detected interferences: hemolysis, lipemia, and icterus (H, L, and I indices).



Goal: ensure that the sample reflects the patient rather than the collection procedure.

Phlebotomy procedure



https://www.youtube.com/shorts/wlh7rOkeVok



Patient position and tourniquet application

Effect of stasis induced by tourniquet application

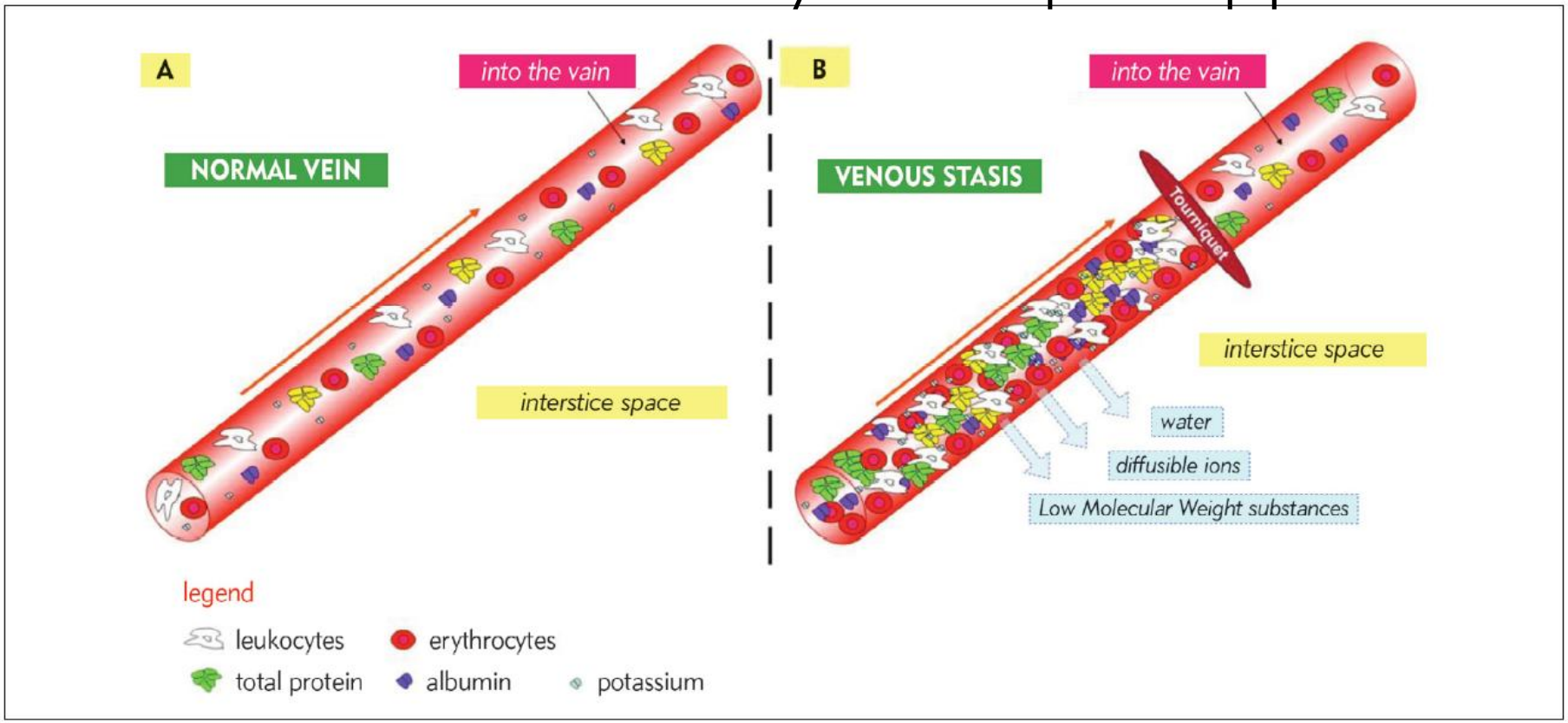
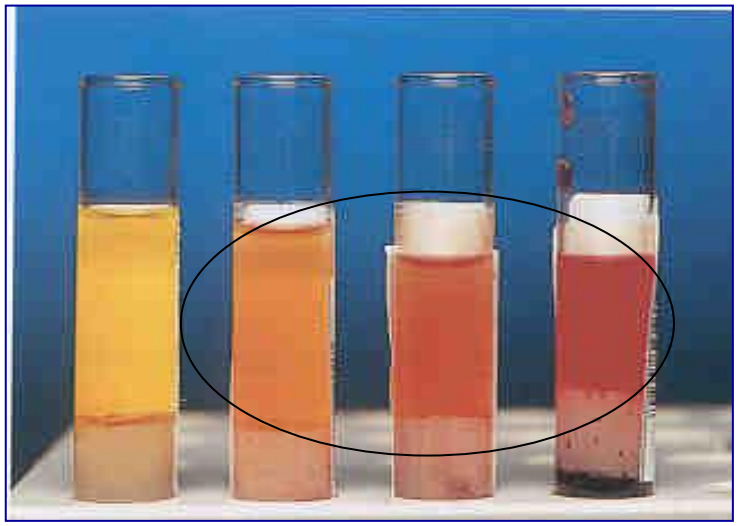


Figure 1 Schematic representation of the effects of stasis induced by tourniquet application on blood constituents. Normal flow of blood through an unobstructed vein (*Figure 1-A*). The obstruction due to tourniquet reduces blood flow thus creating venous stasis (*Figure 1-B*) with net efflux of water from the vessel to the interstice. Elements of low molecular mass diffuse with water whereas high molecular weight compounds and cells concentrate in the vein.

More concentrated sample
lactate

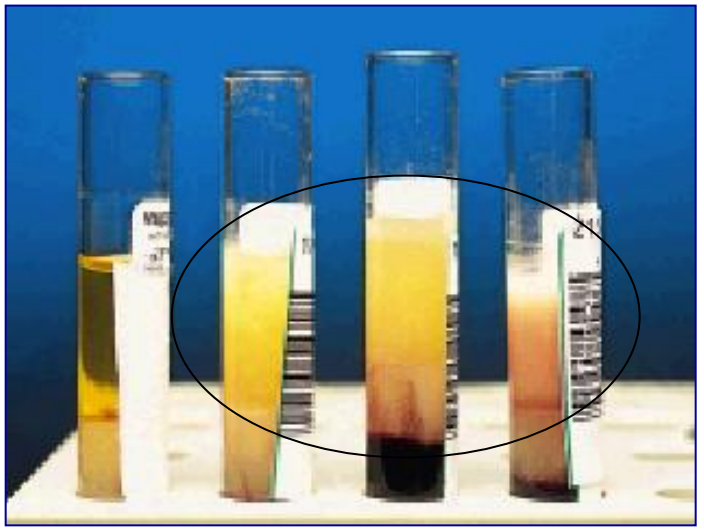
Gabriel Lima-Oliveira y col. Laboratory diagnostics and quality of blood collection. *J Med Biochem 34: 288–294, 2015*

Analytical interferences due to the physical characteristics of the samples



hemolysis

K, LDH



lipemia

Na



fibrin

False negative

Interferent: Factor different to the analyte to be measured that changes the measured results

No visual inspection, we employ indexes

What are the general recomendations for outpatients preparation prior to blood sampling?

1. Fasting state

   12 hours after the last meal.

2. Morning collection

   between 7 and 9 am

3. No confounding factors

   Exercise
   Shift work
   infections
   Jet lag
   Change of time zome

Recommendations for patient preparation for other samples

Special preparation

Centro de Educación al Paciente

Enfermedades
Pruebas diagnósticas
Tratamientos
Cuidados en casa
Consejos de salud

Recogida de orina para la determinación de ácido 5-hidroxiindolacético, catecolaminas, ácido vanilmandélico y metanefrinas

- La determinación de ácido 5-hidroxiindolacético está indicada en el diagnóstico y seguimiento de pacientes con tumores carcinoides.
- La determinación de catecolaminas, ácido vanilmandélico y metanefrinas tiene interés en el estudio de crisis hipertensivas y feocromocitoma.

RECOMENDACIONES DIETÉTICAS

- **Tres días antes de iniciar la recogida de orina y mientras dure la prueba**, debe seguir una dieta normal, a excepción de los siguientes alimentos: plátanos, aguacates, berenjenas, piña, nueces, café, té, bebidas de cola, vainilla y chocolate.

TRACEABILITY



TRACEABILITY

Pacient
Name, last name, medical record
Date of birth/ gender
Hospitalized or outpatient
Doctor, Department

Personnel/Date and time
Request
Extraction
Receptión
Who performed the test
validation

Sample and test
Hemolysis, lipemia, icterus indexes
Comments

Incidences

Results / request
Result and units
Reference values

Analyser

Reagents/Calibrators/Controls
Name
Manufacturer
System
Lot
Expiring date

The barcoded label contains all the information of the process performed to obtain a test result

Tracking and tracing



Have you ever performed a laboratory test?



Point of Care Testing (POCT)



Glucose in capillary blood

Point of Care Testing



Advantages and limitations of POC

FASTER RESULTS

SIMPLER

PORTABILITY

PATIENTS OUTCOMES



INCREASED COST

PERSONNEL: WORK LOAD, LACK OF EXPERTISE

QC

PREANALYTICAL ERRORS IN POCT

Incorrect Patient Identification
Incorrect assignment of results; verify identity using at least two identifiers.

Incorrect Sample Collection Technique
Excessive pressure, inadequate volume, air bubbles, or non-validated samples.

Use of Materials Under Inappropriate Conditions
Expired materials, non-balanced heparin syringes, or improper storage conditions.

Insufficient Staff Training
Errors in collection, handling, and standardized preanalytical procedures.

Delays or Inadequate Conditions Before Analysis
Delays, temperature exposure, and inadequate sample mixing.

Sample Contamination
Incorrect containers, IV-line dilution, or insufficient flushing.

Hemolysis
Improper collection, transport, handling, or delayed processing.

Presence of Clots
Inadequately anticoagulated or poorly mixed samples causing invalid results.

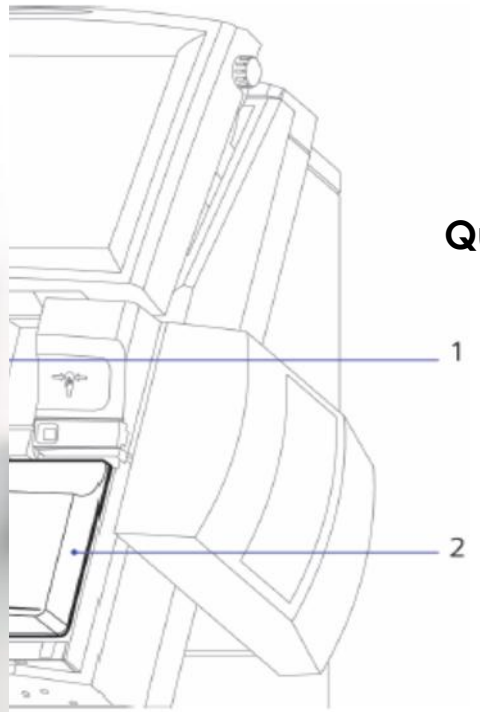
☑ CONCLUSION: Proper patient identification, correct sample collection and handling, appropriate materials, and adequate staff training are essential to prevent most POCT preanalytical errors.

Point of Care Testing

- Urianalysis
- Urine drug tests
- Blood gas analysis
- Capillary glucose
- “Fast” tests for COVID

Blood gas analysis



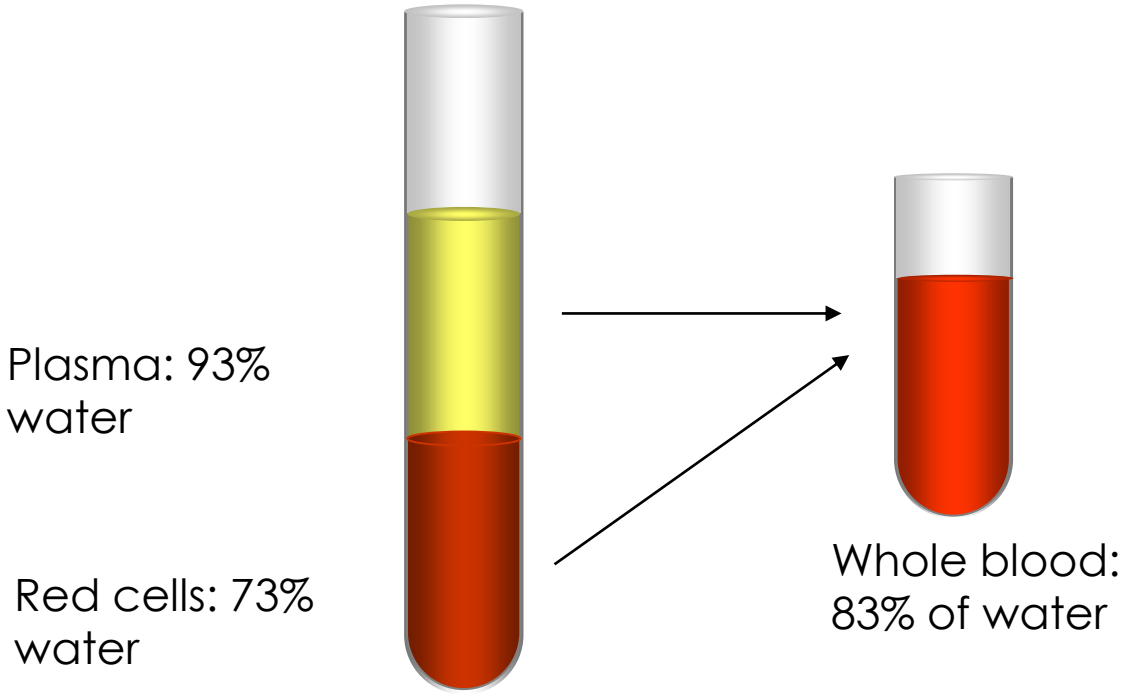


Quality Control Cartridge

Wash/waste cartridge

Whole blood vs plasma

Difference in analyte concentration in whole blood and plasma
Differences are higher when the hematocrit is higher
Different reference values



Water soluble compounds such as glucose or electrolytes are 10% more concentrated in plasma than in whole blood.

SUMMARY

1 Biochemical tests are used in diagnosis, monitoring, screening and prognosis.

2 Core biochemical tests are carried out in every biochemistry laboratory. Specialised tests may be referred to larger departments. All hospitals provide for urgent tests in the ‘emergency laboratory’.

3 Laboratory personnel will readily give advice, based on their knowledge and experience, on the use of the biochemistry laboratory, on the appropriate selection of tests and about the interpretation of results.

SUMMARY

4 Each biochemistry test request should be thought of as a question about the patient and each biochemical result as an answer.

5 Request forms and specimens must be correctly labelled to ensure the correct results for the correct patient can be communicated quickly to the clinician.

6 Many biochemical tests are performed on serum, the supernatant obtained from centrifugation of clotted blood collected into a plain container. Others require plasma, the supernatant obtained when blood is revented from clotting by an anticoagulant. TBL

7 Various preanalytical errors may invalidate results.


CLINICAL LABORATORY
The lab in a hospital
Definition and Purpose
Facilities
Specimen collection area
Others
Operational models
Biological risk
Point of care testing
USE OF THE ANALYSES
Homeostasis assessment
Screening
Diagnosis
Prognosis
Evolution
Monitoring
LABORATORY PROCESS
Preanalytical phase
Test catalog
Test ordering
Patient identification
Specimen and Sample collection
Transportation
Analytical phase
Postanalytical phase
Interpretation of test results
Traceability
SOURCES OF PREANALYTICAL VARIATION
Non-modifiable biological variation
Sex
Age
Race
Pregnancy
Modifiable biological variation
Food intake
Rhythm
Stress
Physical exercise
Drug history
Related to the blood samples
Position
Stasis during venipuncture
Sample collection container
Sample physical characteristics
Stability of constituents
Interferences